Project Report - Day 12

TASK-12: Intent Routing & Agentic Logic Layer

Objective

The objective of this task is to build the **"Brain" of the system** by implementing an **agentic logic layer** that dynamically decides how to respond to user queries.

This involves:

- Understanding user intent
- Routing queries to appropriate logic paths
- Using multiple prompt strategies
- Ensuring safe and controlled outputs

The goal is to transform the system from a static retrieval pipeline into an **intelligent decision-making system**.

Libraries Used

1. Python (Core)

- Used to implement routing logic, control flow, and system integration

2. Custom Modules (Day 11 Integration)

- retriever.py → Used for semantic retrieval from FAISS
- Enables reuse of the data engineering layer

Implementation Summary

1. Designed and implemented an **Intent Router**:
    - Classifies user queries into:
        - Recommendation
        - Trend
        - General
2. Built a **Multi-Prompt System** (prompts.py):
    - Different prompts for different intents
    - Enables dynamic behavior of the system
3. Developed the **Main Pipeline (main.py)**:
    - Integrates:
        - Safety Check
        - Intent Routing
        - Retrieval System
        - Response Generation
4. Implemented **Safety Layer**:
    - Filters harmful or inappropriate queries
    - Prevents unsafe outputs
5. Integrated Day 11 retrieval:
    - Uses FAISS-based semantic search for recommendations

System Design

- **Intent Routing Flow**

User Query → Intent Router → Logic Path Selection

- **Agentic Decision Logic**

Intent	Action
recommendation	Retrieve fashion items using FAISS
trend	Return trend insights
general	Provide generic response

- **Multi-Prompting Strategy**

```
PROMPTS = {
  "recommendation": "Stylist prompt",
  "trend": "Expert explanation prompt",
  "general": "General assistant prompt"
}
```

🔹 Full System Flow

User Query

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Safety Check

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Intent Router

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Prompt Selection

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Retrieval (if needed)

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Response Generation

🧪 Example Outputs

Query 1:

minimalist outfit

Output:

Recommended Outfit:

- Basic black t-shirt

- Black oversized hoodie

- Minimalist beige hoodie

Query 2:

latest fashion trends

Output:

Current trends include oversized streetwear and minimalist fashion.

Query 3:

hello

Output:

I can help you with outfit recommendations and fashion trends.

Errors & Fixes

1. Module Import Errors

- Issue: Python could not find modules across folders
- Solution:
  - Used module execution:
  - python -m Day12_Task.main
  - Updated imports to absolute paths

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2. File Path Issues (FAISS & Metadata)

- Issue: Files not found when running from different directory
- Solution:
  - Used:
  - os.path.dirname(__file__)

    - Ensured correct file paths

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3. Router Import Error

- Issue: ModuleNotFoundError: router
- Solution:
    - Changed to:
    - from Day12_Task.router import IntentRouter

Output Files

- router.py → Intent classification
- prompts.py → Multi-prompt system
- main.py → Final execution pipeline

Conclusion

The Day 12 task successfully implements the **agentic intelligence layer** of the system.

By combining:

- Intent classification
- Dynamic prompt selection
- Integration with semantic retrieval

the system is now capable of making **context-aware decisions** and generating appropriate responses.

This transforms the project into a **complete RAG-based intelligent assistant**, ready for further enhancement and deployment.